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Illustration 9.4: C++Program to find the area of circle

```
#include <iostream>
using namespace std;
int main()
{
        float r, area;
        cout << "\n Enter Radius: ";
        cin >> r;
        area = 3.14 * r * r;
        cout << "\n The Area of the circle is " << area;
}
```

Output:

Enter Radius: 6.5

The Area of the circle is 132.665

In the above example, two variables **r** and **area** are declared as float type in a single statement. Variable **r** is used to hold radius which is a data provided by the user and **area** for holding the area of the circle which is obtained through the formula. For example, if the user inputs the radius value as 6.5, then the statement **area = 3.14 * r * r;** multiplies the value of **r** twice with 3.14 (value of pi) and the resultant value 132.665 will be kept in the variable **area.**

In this case, **r** and **area** both are same type of variables. So, these two variables are declared in a single statement. Declaring the same variables in a single statement reduces the processing time rather than multiple declarations.

(4) double data type:

This is for double precision floating point numbers. (precision means significant numbers after decimal point). The double is also used for handling floating point numbers. But, this type occupies double the space than float type. This means, more fractions can be accommodated in double than in float type. The double is larger and slower than type float. double is used in a similar way as that of float data type.

(5) void data type:

The literal meaning for void is 'empty space'. Here, in C++, the void data type specifies an empty set of values. It is used as a return type for functions that do not return any value.